

Kaggle Competition

Francesco Caperna

July 10, 2026



Agenda

1. **What is Kaggle & how it works?**
2. **Exploratory Data Analysis**
3. **Preprocessing**
4. **Feature engineering**
5. **How to choose the best model?**
6. **Validation Strategy**
7. **Hyperparameter Tuning**
8. **Overfitting**

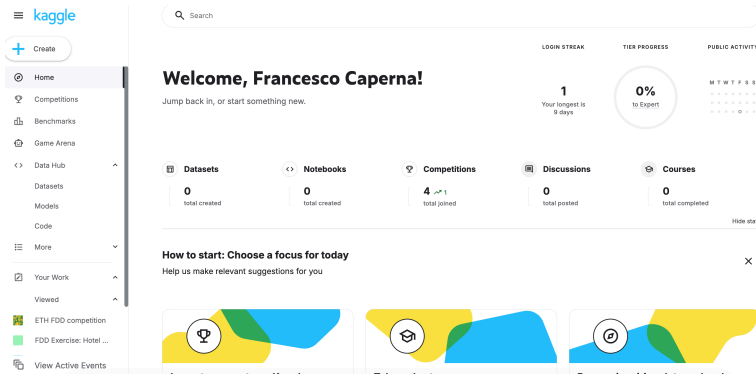
Outline

1. What is Kaggle & how it works?
2. Exploratory Data Analysis
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What is Kaggle?

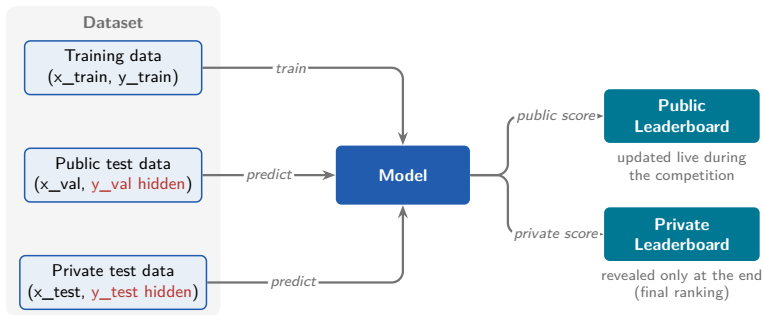
Kaggle is a platform for data science and machine learning:

- **Competitions:** companies post a problem + dataset, you compete to build the best model, ranked on a leaderboard (often with prizes)
- **Datasets:** thousands of free public datasets, ready to download and use



How does a competition work?

#	Team	Members		Score	Entries	Last	Join
1	mlbuarana	 		15.14	98	3d	
2	Jinhuu Yang			25.97	101	15h	
3	Fan of Tong Hui Kang			34.44	67	2d	
4	Team BlackBox	  		33.89	81	12h	
5	Mihai A Vasile			33.89	45	4d	



Why Kaggle?

Real world data are super complicated :(

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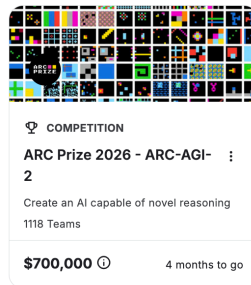
You will never know what is the best model and/or best preprocessing in advance

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Claude Code is really good, but there is a reason if there are professional Kaggle players...



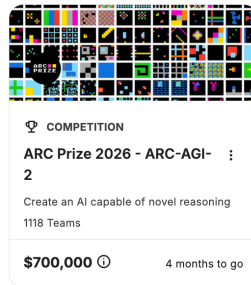
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Machine Learning is like chess, you need to practice...



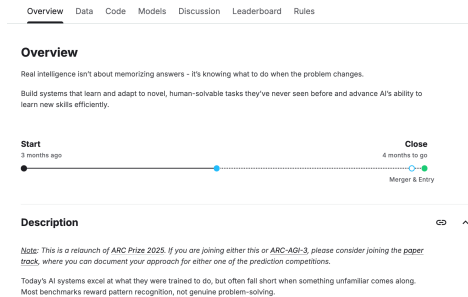
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Exploratory Data Analysis: understanding the problem...

01 What is the problem about?

The **Overview** page of the competition is the first thing to read: goal, timeline, evaluation metric.



The screenshot shows the 'Overview' page of the ARC Prize 2025 competition. At the top is a navigation bar with links: Overview, Data, Code, Models, Discussion, Leaderboard, and Rules. The 'Overview' section contains the following text: 'Real intelligence isn't about memorizing answers - It's knowing what to do when the problem changes. Build systems that learn and adapt to novel, human-solvable tasks they've never seen before and advance AI's ability to learn new skills efficiently.' Below this is a timeline from 'Start' (3 months ago) to 'Close' (4 months to go), with a 'Merger & Entry' point. The 'Description' section includes a note about the competition being a relaunch of the ARC Prize 2025 and a paragraph about AI systems excelling at pattern recognition but struggling with unfamiliar problems.

Overview Data Code Models Discussion Leaderboard Rules

Overview

Real intelligence isn't about memorizing answers - It's knowing what to do when the problem changes.

Build systems that learn and adapt to novel, human-solvable tasks they've never seen before and advance AI's ability to learn new skills efficiently.

Start
3 months ago

Close
4 months to go

Merger & Entry

Description

Note: This is a relaunch of ARC Prize 2025. If you are joining either this or ARC-AGI-3, please consider joining the [paper track](#), where you can document your approach for either one of the prediction competitions.

Today's AI systems excel at what they were trained to do, but often fall short when something unfamiliar comes along. Most benchmarks reward pattern recognition, not genuine problem-solving.

*"To ask the right question is already
half the solution of a problem."*

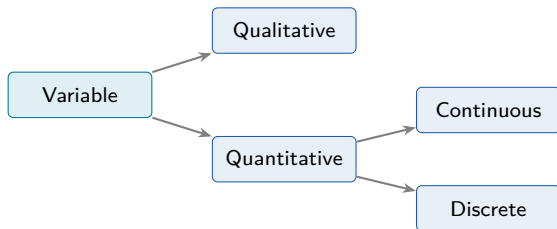
— Carl Jung

Exploratory Data Analysis: understanding the problem...

01 What is the problem about?

02 Dataset size: number of rows and columns

03 Variable types: numerical, categorical, textual, dates, images



Settings Overview **Data** Code Models Discussion Leaderboard Rules Team

X_test.csv (8.53 MB)

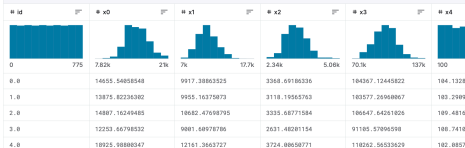


Detail Compact Column

10 of 833 columns

About this file

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Exploratory Data Analysis: understanding the problem...

01 What is the problem about?

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04 Missing values

	feature_1	feature_2	y
x1	2	None	3
x2	None	None	None
x3	5	None	7

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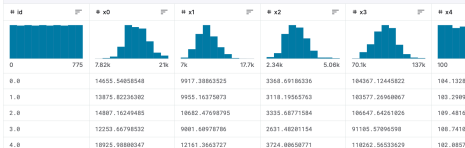


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Exploratory Data Analysis: understanding the problem...

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04 Missing values

05 Outliers

Outliers are unusual values in a dataset: observations that are very different from most of the other values.

	feature_1	feature_2	y
x1	100	10	3
x2	0.1	0.3	5
x3	0.2	0.5	7

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Preprocessing: solving the problem...

01 Missing values: replace with mean, median, and mode

	feature_1	feature_2	feature_3	feature_4	y
x1	5	2	6	3	9
x2	7	None	8	3.5	5
x3	3	4	10	3.5	3
x4	None	None	8	4	2

Preprocessing: solving the problem...

01 Missing values: replace with mean, median, and mode

	feature_1	feature_2	feature_3	feature_4	y
x1	5	2	6	3	9
x2	7	3	8	3.5	5
x3	3	4	10	3.5	3
x4	7.5	3	8	4	2

Preprocessing: solving the problem...

01 Missing values: replace with mean, median, and mode

02 Encoding categorical variables: e.g. one-hot encoding

	feature_1	feature_2	feature_3	feature_4	y
x1	A → [1,0,0]	2	6	3	9
x2	A → [1,0,0]	3	8	3.5	5
x3	B → [0,1,0]	4	10	3.5	3
x4	C → [0,0,1]	3	8	4	2

Preprocessing: solving the problem...

01 Missing values: replace with mean, median, and mode

02 Encoding categorical variables: e.g. one-hot encoding

03 Scaling variables: e.g. Standard scaler

	feature_1	feature_2	feature_3	feature_4
μ (mean)	4.375	3.0	8.0	3.5
σ (std)	3.0695	0.7071	1.4142	0.3536

Standard scaler:
$$x' = \frac{x - \mu}{\sigma}$$

	feature_1	feature_2	feature_3	feature_4	y
x1	0	2	6	3	9
x2	7	3	8	3.5	5
x3	3	4	10	3.5	3
x4	7.5	3	8	4	2

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Standard scaler:
$$x' = \frac{x - \mu}{\sigma}$$

	feature_1	feature_2	feature_3	feature_4	y
x1	-1.4	-1.4	-1.4	-1.4	9
x2	0.8	0	0	0	5
x3	-0.4	1.4	1.4	0	3
x4	1	0	0	1.4	2

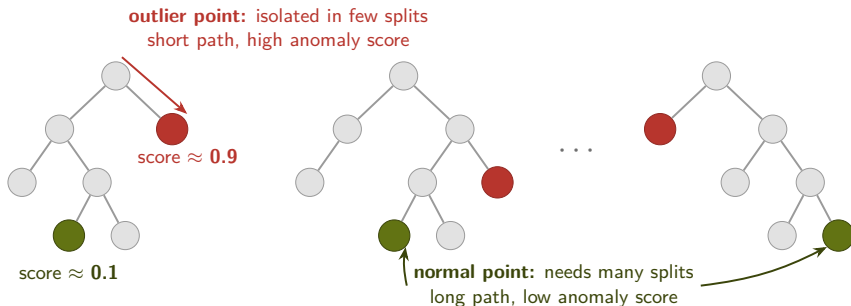
Preprocessing: solving the problem...

01 Missing values: replace with mean, median, and mode

02 Encoding categorical variables: e.g. one-hot encoding

03 Scaling variables: e.g. Standard scaler, Robust scaler

04 Outliers: e.g. Isolation Forest



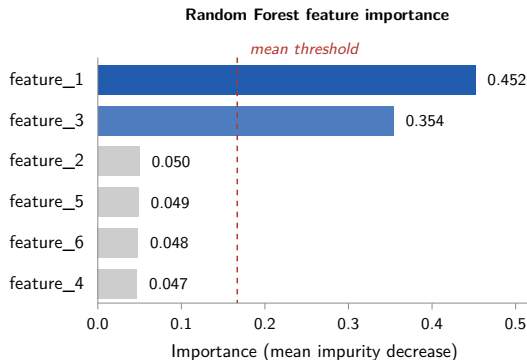
the anomaly score is the average path length over all random trees

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Feature engineering: solving the problem...

01 Select most relevant features: e.g. feature selection with Random Forest



Feature engineering: solving the problem...

01 Select most relevant features: e.g. feature selection with Random Forest

02 Combining features: e.g. summing up two features

	feature_1	feature_2	feature_1 + feature_2	feature_1 · feature_2	y
x1	-1.4	-1.4	-2.8	1.96	9
x2	0.8	0	0.8	0	5
x3	-0.4	1.4	1	-0.56	3
x4	1	0	1	0	2

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Model to use: understanding the problem first...

- It depends on the data

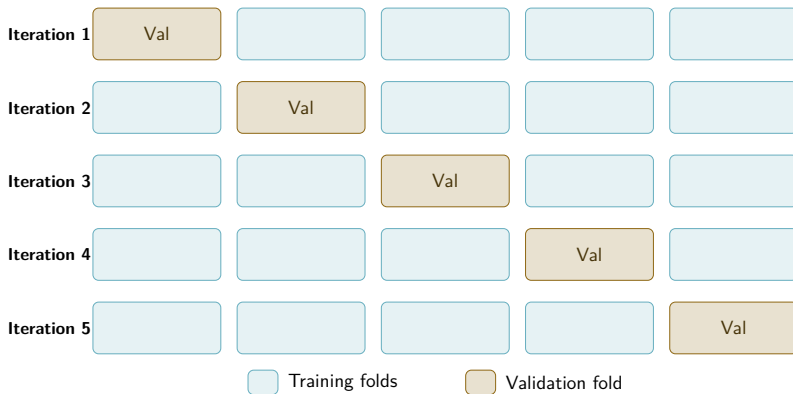
Type of data	Model to use
Tabular / structured	Gradient boosting (XGBoost, LightGBM, CatBoost), Random Forest
Images	CNN, Vision Transformers (ViT), Diffusion models
Text / NLP	Transformers (BERT, GPT)
Audio / speech	CNN on spectrograms, Transformers (wav2vec, Whisper)
Graph / network	Graph Neural Networks (GNN)
Small dataset	Simple models: linear/logistic regression, SVM, shallow trees

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Validation Strategy: solving the problem...

- K-fold cross validation: each fold is used once for validation, the final score is the average over the K iterations

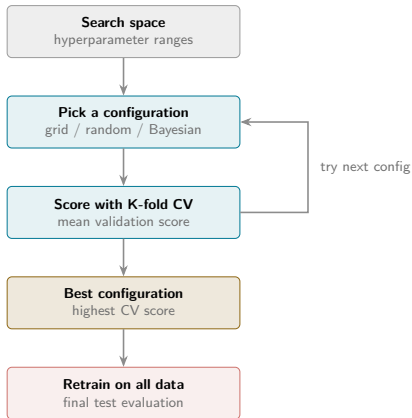


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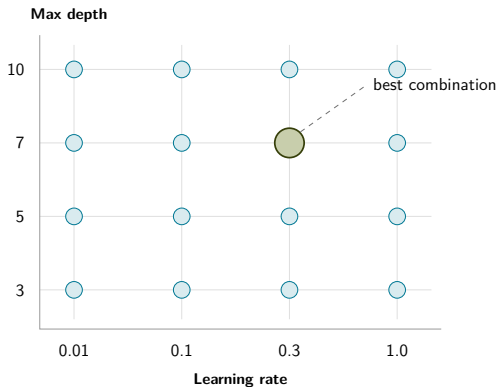
Hyperparameter Tuning: solving the problem...

- Search the hyperparameter space iteratively: every configuration is scored with cross validation, the best one is kept



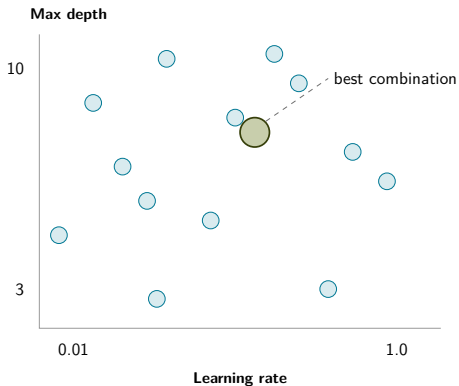
Hyperparameter Tuning: Grid search

- Tries all combinations on a fixed grid: exhaustive but expensive, the cost explodes with the number of hyperparameters



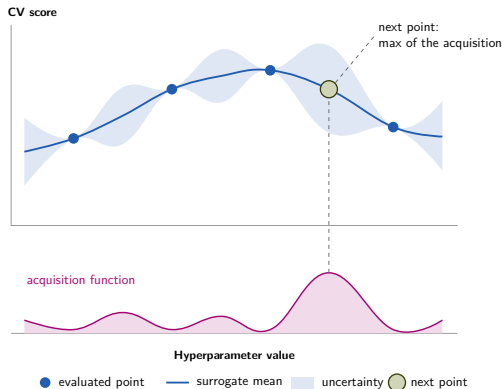
Hyperparameter Tuning: Random search

- Samples random combinations: often finds a good configuration with far fewer trials than grid search



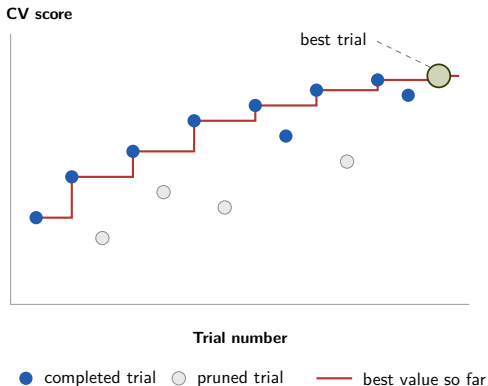
Hyperparameter Tuning: Bayesian optimisation

- Uses the results seen so far to choose the most promising configuration to try next, balancing exploration and exploitation



Hyperparameter Tuning: Optuna

- Optuna: Python framework that implements Bayesian optimisation in practice, with some modifications such as pruning unpromising trials early



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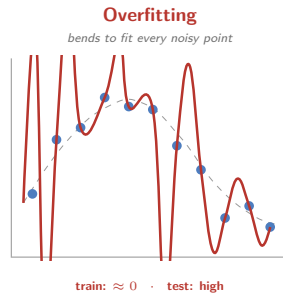
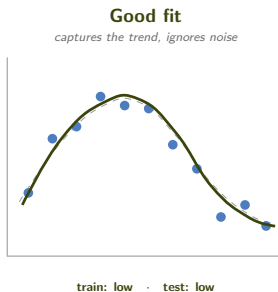
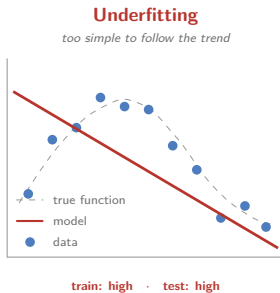
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Overfitting... be careful about your result

Model too complex

How to avoid it?

1. Cross validation
2. Feature selection

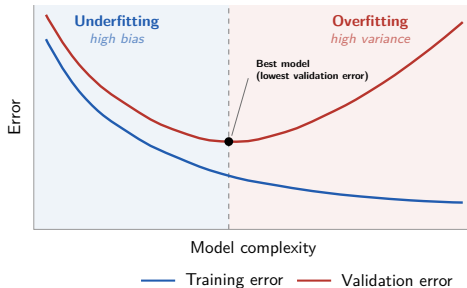


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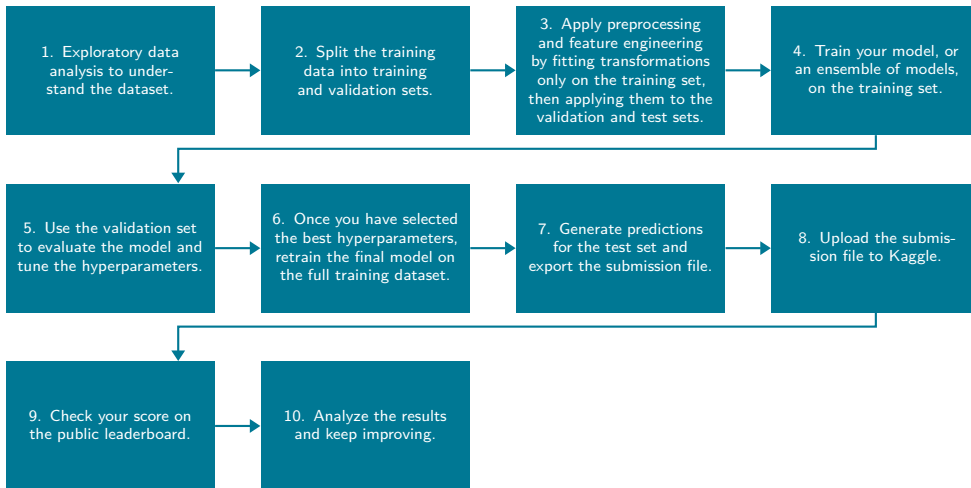
Model too complex

How to avoid it?

1. Cross validation
2. Feature selection
3. Early stopping
4. Regularization techniques



Workflow for a Kaggle competition



Thank you